

Advanced Technique-Based Combination of Innovation Education and Safety Education in Higher Education

Deng-Guang Yu,* Qiang Li, Wenliang Song, Lin Xu, Kaihui Zhang, and Tao Zhou*



Cite This: *J. Chem. Educ.* 2023, 100, 507–516



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ABSTRACT: The essence of higher education is to cultivate innovative talents for the society. Hence, higher education institutions (HEIs), on the one hand, need to respond to social needs and government policies. On the other hand, they need to keep pace with the times and improve the details of teaching to improve the effectiveness of education. In this paper, several combination education examples in HEIs about safety education and innovation education are shown to encourage students to realize innovations in technique during their laboratory experiments on electrospinning. To obtain professional knowledge and traditional professional skills, students are taught to implement electrospinning, an advanced technique for treating polymers, in the research laboratory. Safety education is also taught, during which the students are encouraged to play their subjective roles and ponder on how to apply innovations for safe, effective, and innovative implementation. The students apply a series of small technological innovations for the spinneret, which is the most innovative part of an electrospinning system. They can modify the spinneret for safer, more energy-saving, and more convenient operations of the single-fluid blending, coaxial, triaxial, and side-by-side electrospinning processes. Moreover, the innovations can deepen their professional knowledge of electrospinning and stimulate their enjoyment of innovations. Safety education on electrospinning can be a powerful tool for promoting technological innovations in students and increasing their awareness of related safety issues. The reported protocols in this paper show a new way for conducting combined education in HEIs on advanced techniques.

KEYWORDS: Laboratory Experiments, Safety Education, Materials Science, Technique Innovation, Nanomaterials, Polymers, Electrospinning



■ FROM KNOWING THE WHAT-WHY-HOW OF PROFESSIONAL TECHNIQUES FOR STUDENTS TO GRASP THEM

Higher education institutions (HEIs) in China that implement professional education are mainly colleges, universities, some scientific research institutes, and specialized and vocational education schools. In these HEIs, students commonly have considerable knowledge of the professional techniques that the teachers have taught them in the classroom.¹ They often have solid theoretical knowledge and understanding of the operation procedures and potential applications of those professional techniques. However, their practical skills and ability to integrate theory with practice often do not meet the requirements of the field they work in after graduation.² Thus, practical and innovative education is often one of the most difficult challenges, particularly in HEIs that do have not enough practical teaching resources. Professional laboratory experiments are useful sources to be adapted for providing potential solutions.

The situation could be attributed to several possible reasons: (1) Professional education in many HEIs cultivates all kinds of professional talents at all levels after basic education, but the

main courses always aim to improve students' theoretical knowledge; (2) the HEIs do not have enough staff members with extensive first-hand experience with the professional techniques or practical conditions for the students to improve their professional skills; and (3) some of the advanced professional techniques are taught discretely, unsystematically and integrated into in situ lecture for provoking innovations directly from professional practices.³

Whether graduate, undergraduate, or vocational education, the final objective is to provide potential workers and technicians with professional skills and technical know-how to serve the advancement of society. However, nurturing the students to become well-educated professionals and technicians is never easy. Students can grasp an advanced professional technique completely by learning the “whats”

Received: June 9, 2022

Revised: October 18, 2022

Published: January 5, 2023



and “whys” in classrooms. However, most important and most difficult segment is the “hows” and their hands-on learning, particularly for some advanced professional techniques such as electrospinning and electrospraying (Figure 1). Intrinsic motivation, for example, their safety, can be explored for effective practice education, which is always positively correlated with their study performances.⁴



Figure 1. Safety education is the core content of lessons on advanced professional techniques and guarantees professional innovations.

Only after knowing how to effectively implement the techniques can the students further cultivate their capabilities to integrate theory with practice and develop their innovative capability for improving their professional skills.^{5,6} Later, based on their practical experience with innovations, students will know how to deepen their knowledge of new techniques intentionally and become self-improving talents. This virtuous circle requires the integration and spiral rise of theory and practice and consequently continuous knowledge restructuring and safe practices.^{7,8} Allowing these things to take place in lab experiments can benefit the students' lifelong learning and innovation abilities.

For professional techniques associated with materials, chemistry, and many applied scientific fields, safety is one of the most important issues,^{9,10} which may be ignored during lessons on theories in classrooms and can be taken for granted easily, being replaced by general safety knowledge.¹¹ Many professional techniques have their special features related to their safe implementations, which serve not only as the core section of a successful process but also as a stimulus point for innovations.⁴ In the spiraling process of “know-how,” safety is the foundation for commencing practice on a professional technique and learning how to optimize the experimental parameters, being able to distinguish the subtle differences during the working process, enriching all related experiences, and realizing innovations (Figure 1).

This article uses electrospinning, which is an advanced electrohydrodynamic atomization (EHDA) method similar to electrospraying, as a model professional technique.¹² This work introduces what electrospinning is and why it is popular for fabricating nanomaterials for a wide variety of functional applications.^{13,14} Its special features on safe implementations and the related dangerous issues are later described, which are interesting materials for teaching students at different levels. As

a source of inspiring innovations, the safe operations for preventing possible risks during the electrospinning process are related closely to the electrospinning instrument, particularly its core part, namely, the spinneret.¹⁵ Scientific instruments have long been a vital part of science, paving the pathway to remarkable scientific advancements and provoking students' interest and motivation toward practical learning and innovations.¹⁶ A series of related innovations from the students are exhibited. These innovations, which result from the promotion of safety education of the students in the laboratory, include how to implement (1) single-fluid blending electrospinning by using a Teflon-coated metal spinneret,¹⁷ (2) coaxial and triaxial electrospinning using a polymer-metal-combined spinneret,^{18,19} (3) side-by-side electrospinning through Teflon-coated parallel metal capillaries,²⁰ and (4) the collection of samples for transmission electron microscopy (TEM) and aligned nanofibers while avoiding high voltage.¹⁶ How these innovations have successfully deepened students' cognition of electrospinning is also depicted.

■ KNOWLEDGE OF THE “WHAT-WHY-HOW” OF ELECTROSPINNING

The concept of electrospinning is not difficult for students majoring in materials science and engineering or those involved in polymer, nanomaterials, and nanotechnologies.^{21,22} Electrospinning was originally referred to as electrostatic spinning, which combines traditional wet spinning and dry spinning. It was originally used to prepare nonwoven mats and is a technique and concept in textiles.²³ Similar to electrospraying, it has the advantage of providing easy interactions between electrostatic energy and liquids to solidify treated working fluids into solid nanofibers.²⁴ The theory is difficult because it involves several overlapping disciplines, such as hydrodynamics, electrokinetics, polymer rheology, and so on. Thus, no working mathematics models can be effectively utilized to treat all kinds of electrospinnable working fluids. However, this case has little influence on the popularity of electrospinning in creating thousands of interesting functional nanofibers in laboratories all over the world.²⁵ Thus, it is a good model to be selected as an advanced technique for conducting laboratory education. Education of laboratory practices based on a new technique (e.g., paper microfluidic technology²⁶) not only can train the students' process skills^{27,28} but also can strengthen their collaborative and scientific humanistic spirit.^{29,30} Furthermore, an effective combination of classroom education with the related laboratory practices can provide a review on the knowledge learned in class, cultivate the innovative consciousness of integrating theory with practice, and promote integrated multidimensional learning.^{31,32} In this study, the laboratory practice and safety education associated with electrospinning is exhibited, which is aimed to (1) deepen and broaden the students' fundamental knowledge about electrospinning learned in the classroom courses; (2) cultivate students' practical skills and the relevant safety awareness; and (3) explain those innovations on electrospinning systems that resulted from the considerations of safe implementations for fostering the students' innovation consciousness and innovation ability. The assessments on the goal achievement were conducted according to students' usual performances (two of the seven times were in laboratory) and the final test scores. The results demonstrated that the teaching purposes were fully realized to an excellent standard.

Why is electrospinning popular even without a certain support theory? One important reason can be derived from its development history. Electrospinning was first patented in 1934 but was not used for several decades.³³ The 1990s saw the emergence of the nano era. Adapting to the times, Reneker revived this technique as a useful method for preparing nanoproducts, such as one-dimensional (1D) nanofibers in 1995 (in contrast to 0D nanoparticles and 2D nanosheets such as graphene).³⁴ From then on, reports about electrospinning soared drastically. The second reason is that a series of advanced material characterization methods such as scanning electron microscopy (SEM) and transmission electron microscopy (TEM) have been developed for assessing electrospun nanofibers.^{17–20} The most important reason has a close relationship with the advantages of the working process and the resultant nanoproducts. The process advantages of electrospinning include the following: (1) a cheap and easy-to-assemble working system; (2) a straightforward one-step production process; and (3) easy scale-up for large-scale production although it is mainly a laboratory technique.^{35,36} Electrospun nanofibers have the following well-known properties: (1) unique properties, such as a small diameter, large surface, and high porosity; (2) it is easy to be utilized in creating nanoproducts of inorganic and ceramic materials with nanofibers as templates; (3) the ability to be endowed with various complicated nanostructures such as core-shell, Janus, trilayer core-shell, and their combinations through the structured spinneret in a top-down manner;^{37,38} and (4) can be used as building blocks for constructing novel bulk materials or as reinforcement materials for achieving super mechanical performances. Thus, electrospun nanofibers are demonstrated to have potential applications in almost all kinds of scientific fields.^{39,40} The developing trends and the mainly related fields are included in [Supporting Information](#).

The “what” and “why” of electrospinning can be taught to students in the classroom, along with the elements of electrospinning equipment. [Figure 2a](#) shows an electrospinning

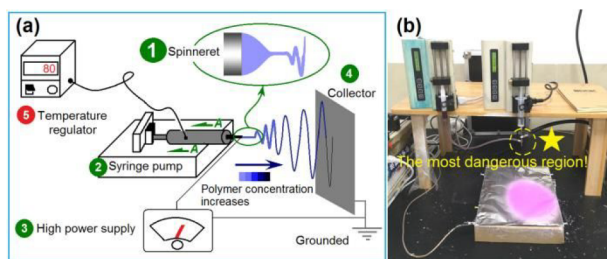


Figure 2. Electrospinning system: (a) a diagram of evaluated temperature electrospinning; (b) an image of an ongoing coaxial process.

system with auxiliary equipment to provide elevated temperature (part 5). The other four parts are the spinneret (1), syringe pump (2), high-power supply (3), and collector (4). All these elements are organized to manipulate the interactions of electrostatic energy and working fluids. Details on how to set up electrospinning with these elements are given in [Supporting Information](#).

However, how the electrospinning process is implemented may puzzle the students. Despite it being a single-step process, its successful implementation requires many minor tricks. Safety is the most important issue for beginners in starting

their experiments on preparing polymeric nanofibers. [Figure 2b](#) shows an image of coaxial electrospinning in progress. The students are taught how to manipulate the pumps to provide suitable flow rates, adjust the high-power supply to apply a high voltage, determine the fiber deposition distance, and ensure that these parameters match to create high-quality nanofibers. Safety issues can arise during any of these steps. The most dangerous region with unique safety issues is always related to the spinneret, which is the convergence point for all working fluids and the electrostatic energy (marked by a star in [Figure 2b](#)). The most frequent safety issue is electric shocks. Thus, safety education about electrospinning concentrates on the spinneret, thereby generating several technique innovations. The operations for teaching the students to implement coaxial electrospinning are given in [Supporting Information](#).

■ SAFETY ISSUES DURING ELECTROSPINNING FOR EDUCATION

The common procedures in industrial production and materials fabrication in a laboratory are similar in that they start from the raw materials and terminate at the desired functional products, and a series of intermediate technical processes are involved for material conversions. Two types of dangerous issues are associated with electrospinning. The first is general safety issues, which are similar to the issues that occur during other chemical or physical procedures for treating materials, and the other is the special safety issues associated with the electrospinning process. Students often prefer information related directly to the target experiments.⁴¹ Their personally experienced safety issues, whether general or special, will augment the learning effects. In this study, the laboratory safety education for the students (both undergraduate and postgraduate) was implemented according to the following three gradual deepening steps: (1) learn about the fundamental knowledge of electrospinning (including knowledge about laboratory safety) based on the open course of “Preparation of nanofibers and potential business opportunities” in University of Shanghai for Science and Technology (Lecturer: Deng-Guang Yu); (2) receive the safety education on the electrospinning apparatuses in the scientific research laboratory after half a semester of theoretical study; (3) install the electrospinning system and implement an electrospinning process for experiencing safety issues, thinking of innovations, and preparing polymeric nanofibers, which lasts for 4 weeks for all the students. Later, the courses are organized in an open manner, the students are encouraged to propose ideas, organize scientific research activities, think about innovative contents, and even write reports and papers. The learning effects of all the students will be evaluated through a combination of their ordinary classroom and laboratory performances and the achieved scores of a post survey ([Supporting Information](#)).

Whole-Process Safety Issue

[Figure 3](#) is a whole process for conducting a series of experiments for preparing and characterizing electrospun nanofibers. The first step is to prepare the working fluids based on the designed products, which should have fine electrospinnability.⁴² Although electrospinning is initially a polymeric nanofiber creation method, other molecules, such as surfactants, salts, chemical molecules, and lipids, can be codissolved into the working fluids to create nanocomposites and nanohybrids. Then, the electrospinning process is

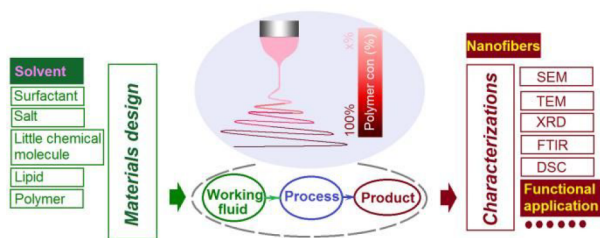


Figure 3. Whole process for converting raw materials into nanofibers.

optimized to convert the working fluids into solid nanofibers as the second step.⁴³ During the third step, the resultant nanofibers undergo a series of characterizations of morphologies (such as SEM and TEM), physical state, compatibility [such as X-ray diffraction (XRD), Fourier transform infrared spectra (FTIR), and differential scanning calorimetry (DSC)], and some analyses of their functional performances.

The first and third steps involve many common safety issues similar to those in other techniques, such as toxicity of the poisoning organic solvents and their flammability and explosive properties during the working fluid preparation process. Exposure to ionizing radiation and X-ray radiation is also possible during the analysis processes using TEM, SEM, and XRD. These issues will not be discussed here. The key special safety issues are those for the safe implementation of electrospinning. These special safety issues can be more intelligible when unprofessional analogies are utilized, which have a rich history in chemistry education to help students visualize a new concept.⁴⁴

Special Safety Issues for Safe Implementation of Electrospinning

The successful operation of electrospinning should be characterized not only by the formation of solid nanofiber but also by the successful installation and initiation, stable and robust operation, and smooth termination of the entire electrospinning process. During these procedures, the safety issues are associated closely with two aspects: electro and hydro (Figure 4). Electric shocks, injury, poisoning, burning, and fire are all possible risks during an electrospinning process. The severity of these issues is presented in Table 1 and is

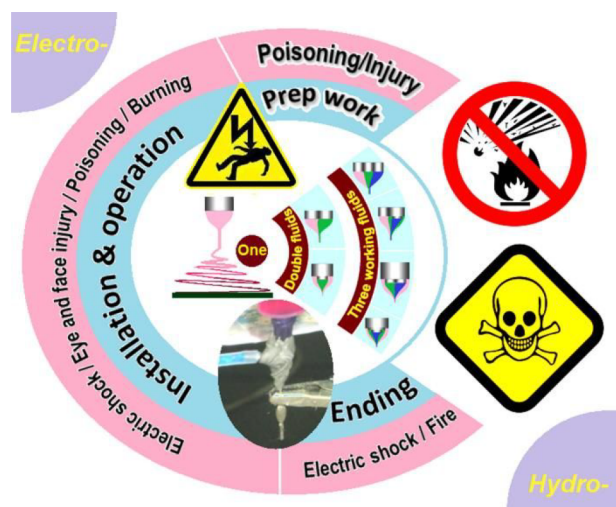


Figure 4. Special safety issues for the safe implementation of electrospinning always focus on the key element, i.e., the spinneret.

Table 1. Potential Safety Issues during Electrospinning

Working process		Main safety problems	Possible reasons	Grade ^a	
Preparing fluids		Poisoning, fire, and explosion	Organic solvents	★	
Installation & operation	Installation	Electric shock	Early start power supply	★★★	
		Eye or face injury	Filling working fluid	★★	
		Fire	Short circuit	★	
	Operation	Electric shock	Carelessness/electric leakage	★★★	
		Poisoning	Organic solvent atmosphere	★★	
		Fire	Short circuit/poor contact	★★	
		Eye or face injury	Clogging	★	
		Damage the instrument	Operation error	★	
		Ending	Electric shock	Carelessness/forget to turn off	★★
			Fire	External open fire	★
			Damage the power supply	Return to zero	★
Applications		Poisoning	Residual solvent	★	

^aThe number of stars reflects the frequency and severity.

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denoted by stars. The most severe and most frequent risk is electric shock. Thus, a wide variety of measures have been taken in many electrospinning laboratories to prevent electric shocks. Related innovations after safety education are centered correspondingly around the prevention of electric shocks.

Possible electric shock may cause students to (1) become afraid of conducting experiments and even of approaching the electrospinning system, (2) try their best to perform electrospinning carefully, and (3) think of how the equipment, particularly the spinneret, can be improved to avoid electric shocks. During laboratory experiments, electric shocks can occur during all the procedures, such as during the installation and initiation of electrospinning (e.g., wrong operation of first switching on the power supply and later adjusting the flow rate of working fluids); the running process of electrospinning (e.g., touching the metal spinneret out of carelessness and removing a clogged semisolid substance at the nozzle of the spinneret using the naked hand); and the termination of electrospinning (e.g., not switching off the power supply before removing the syringe and spinneret due to carelessness).

Aside from electric shocks, other risk factors are burning and explosion, trauma, injury, and damage to the instruments, which all happen occasionally (Table 1 and Figure 4). Burning and explosion risks should be avoided during elevated-temperature solution (due to organic solvents such as methanol, ethanol, acetone, and chloroform) or melt electrospinning. Instrument damage may result from a short circuit due to the loose installation of the spinneret on the syringe and falling off of the spinneret to the collector. Eye or face injury or trauma can occur during the filling of working fluids and from the metal spinneret and alligator clip.

TECHNIQUE INNOVATIONS OF STUDENTS FROM THEIR SAFETY EDUCATION

From Safety Education to a New Polymer-Metal-Combined Monoaxial Spinneret and New Cognition about Electrospinning

In the field of electrospinning, the most common spinneret is the monoaxial spinneret, which is essentially a metal capillary in most situations.⁴⁵ The most common issues are electric shocks. However, fire and burning also occur frequently due to (1) flammable organic solvent evaporation, (2) flammable electrospun nanofibers, and (3) sparks that are easily caused by poor contact. Figure 5a shows an image of a burned concentric

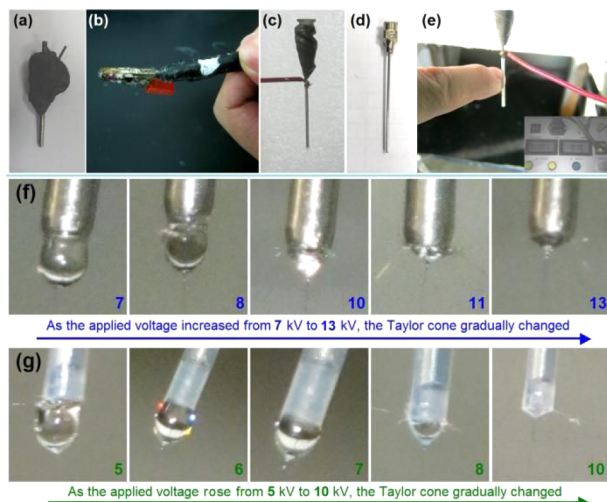


Figure 5. Teflon-coated spinneret for blending electrospinning: (a) digital picture of a burned spinneret; (b) rusty alligator clip; (c) Teflon-coated spinneret; (d) whole-metal spinneret; (e) index finger touching the spinneret of the Teflon-coated section safely during the working process; the bottom-right inset shows the applied voltage of 10.1 kV; (f) electrospinning process using a whole-metal spinneret; (g) electrospinning process using a Teflon-coated spinneret.¹⁷

spinneret, with the metal capillaries and the carbonized epoxy resin being the only remaining materials, caused by flying nanofibers and working fluids. Figure 5b shows a rusty alligator clip that can lead to poor contact and easily create sparks in a drying atmosphere. Enlightened by safety education, the students become more careful during electrospinning experiments and can conceive innovations of the spinneret based on the analogies of routine daily life for electrical safety.

Figure 5c shows a section of Teflon tubing coated directly on a metal capillary as an electric cable, leaving only a small metal section for the copper lines to wrap to convert electric energy.¹⁷ Unlike the whole-metal spinneret in Figure 5d, most of the metal spinneret was covered by insulating polymers. Thus, it is not strange that a person might not feel anything when an index finger touches the surface of the Teflon tubing covering the metal capillary (Figure 5e), where the working high voltage is 10.1 kV (the bottom-right inset of Figure 5e). This minor innovation of a Teflon-coated monoaxial spinneret increases the safety of implementing electrospinning.¹⁹ Meanwhile, the innovation of the Teflon-coated spinneret has the following benefits aside from ensuring safe operation:

- (1) Compared with the electrospinning process that uses a whole-metal spinneret, the electrospinning process that

uses a Teflon-coated spinneret requires a smaller applied high voltage to initiate the working processes, as shown in Figure 5f,g. The voltages for stable and robust electrospinning without dripping droplets of the working fluids are 10 and 8 kV, respectively. Thus, the Teflon-coated spinneret can increase the safety from two angles because of its insulated property and because it can effectively reduce the applied voltage value for avoiding electric shocks.

- (2) The Teflon-coated electrospinning process is more energy-saving than the metal capillary-based electrospinning process. The insulated property of Teflon can effectively prevent the dispersion of electrostatic energy to the environments from the surface of the spinneret.⁴⁵
- (3) In almost all reports about electrospinning, the spinneret is inevitably made of metal because of its fine conductive property. However, the involvement of Teflon as a part of the spinneret clarifies to the students that it is the working fluids that require the electrostatic energy to commence the electrohydrodynamic processes. As for how energy can be converted to the working fluids, the metal section can make it possible, and a combination of metal and polymer for spinneret can support a better working process.

From Safety Education to Innovations of Coaxial/Triaxial Electrospinning

Coaxial electrospinning processes are different from single-fluid blending electrospinning in the following aspects: (1) a concentric spinneret, which is still the most dangerous place for operation; (2) two working fluids, which are involved in the single-step preparation; and (3) most importantly, a core-shell nanostructure can be produced, which was once considered the most important breakthrough in this field.⁴⁶

Figure 6a shows a digital picture of the commercial concentric spinneret, which has two inlets and a concentric outlet at the nozzle of the spinneret but all are made of stainless steel for good conductivity. The whole-metal spinneret means (1) a larger place for possible contact with the operator, thereby creating a severe safety issue and (2) a larger surface for dispersing electrostatic energy to the environment. Based on safety education, the students innovatively conceived two kinds of compound concentric spinnerets, which are composed of both metal capillaries and plastic tubing.

The digital pictures in Figure 6(b1–b4) show a new kind of combined concentric spinneret that comprises one metal capillary and plastic tubing.¹⁸ An additional needle with high elastic silicone tubing was utilized to transfer the sheath working fluid. The advantages of this spinneret are its more convenient operation for commencing the working process (the commercial spinneret in Figure 6a makes arranging two syringe pumps difficult) and for washing the spinneret after finishing the preparation, aside from ensuring a safer and more energy-saving performance.

Similar to the Teflon-coated monoaxial spinneret, the concentric spinneret can be built using Teflon tubing.⁴⁷ Figure 6c shows another kind of concentric spinneret. The digital photos in Figure 6(c1–c4) show the whole image of the spinneret, its outlets, a slight protrusion of the inner metal capillary, and connections with two syringes, respectively. During all preparations of the concentric spinneret, attention is always paid to the concentricity of the two passages that lead

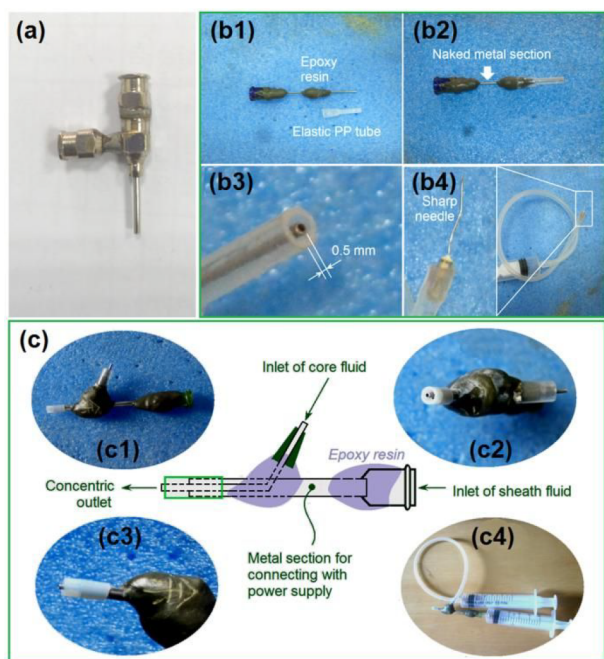


Figure 6. Teflon-coated concentric spinneret for coaxial electrospinning: (a) digital picture of the commercial concentric spinneret; (b1–b4) a new kind of concentric spinneret that combines one metal capillary and plastic tubing;¹⁸ (c) another kind of concentric spinneret, (c1–c4) are digital photos of the whole image of the spinneret and outlets; slight projection of the inner metal capillary and connections with two syringes. Reprinted with permission from ref 47, copyright 2022, The Author(s).

the sheath and core working fluids, i.e., the two capillaries of the core and sheath sections have a common axis or are coaxial, which is the reason for coaxial electrospinning. However, the prepared nanofibers, which were formed after several thousand rounds of electrical drawing and elongations, cannot be ensured to have a common axis. Hence, the right term for them should be “core-shell” or “core-sheath.” This condition makes the students understand more clearly that “coaxial nanofiber” is not an appropriate term for the resultant nanofibers, although it frequently appears in the literature.

Inspired by the combined coaxial spinneret in Figure 6b and safety education, the students created their innovations for the trilayer concentric spinneret and the trilayer coaxial electrospinning,¹⁹ Figure 7 shows the combined trilayer coaxial

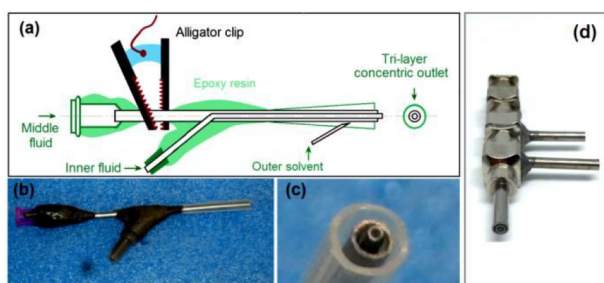


Figure 7. Triaxial electrospinning: (a) a diagram of the organization of a triaxial concentric spinneret; (b,c) digital pictures of the whole image and trilayer concentric outlets of the spinneret;¹⁹ (d) a digital photo of the commercial triaxial spinneret.

spinneret. Figure 7a is a diagram of the organization of a triaxial concentric spinneret. Figure 7b,c are digital pictures of the whole image and trilayer concentric outlets of the spinneret, respectively. Compared with a commercial triaxial spinneret (Figure 7d), which has a large surface that presents hidden risks and energy waste, the innovative spinneret leaves only a small metal section for the alligator clip to convert the electrostatic energy to the working fluids, thereby ensuring safety and energy-saving.

From Safety Education to Innovations of Side-by-Side Electrospinning

Side-by-side electrospinning is one of the most difficult challenges in the field of electrospinning. In this process, two different working fluids are guided to a side-by-side spinneret (Figure 8a). Similar to coaxial electrospinning, the side-by-side

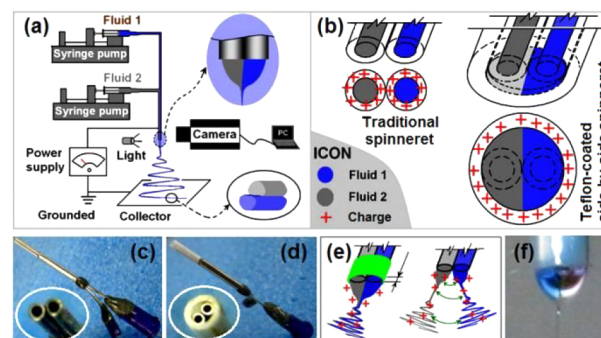


Figure 8. Teflon-coated side-by-side spinneret for side-by-side electrospinning: (a) a diagram showing side-by-side electrospinning for preparing Janus nanofibers; (b) charge distributions at the nozzle of the spinnerets; (c) a digital picture of the traditional parallel metal-capillary spinneret; (d) a digital picture of the Teflon-coated side-by-side spinneret; (e) a diagram of the performances of the two kinds of side-by-side spinneret; (f) a digital picture of the compound side-by-side Taylor cone, reproduced with permission from ref 20, copyright 2015, Elsevier.

electrospinning system has four parts. The only difference lies in the spinneret. In coaxial electrospinning, a concentric spinneret with a capillary is nested into another capillary to form the spinneret, which also serves as a template for creating core-shell nanofibers. According to this configuration, it is easy to conceive of two parallel capillaries bounded together for preparing a side-by-side spinneret, similar to that shown in the diagram in Figure 8b and the digital picture in Figure 8c (the inset shows the side-by-side outlet).²⁰ However, this configuration inevitably leads to a failure in generating nanofibers with integrated Janus structures. The two working fluids bear the same charges and only contact at one point when they are pumped out of the nozzle. “As electric charges repel, unlike ones attract” is the natural rule by which the two working fluids will mutually repel during the electrical drawing processes (Figure 8e). The very limited publications on Janus nanofibers from the parallel capillary spinneret are due to the high concentrations of polymers and the related high viscosity and high compatibility. The adhesive forces overcome the repelling forces.

Similar to the Teflon-coated monoaxial spinneret, a Teflon tube can be used to coat the parallel side-by-side spinneret,²⁰ which is originally an idea from safety education on preventing possible electrical shock. This simple trick represents a useful innovation and generates satisfactory results. A diagram and a

digital picture of the Teflon-coated side-by-side spinneret are shown in Figure 8b,d (their insets show the side-by-side outlets), respectively.

This innovation and the related results for generating integrated Janus nanofibers further deepen the students' cognition of electrospinning in the following aspects: (1) The outlet of the spinneret nozzle should have a full circle for the electrical energy to enable it to perform its important role in forming the compound Taylor cone; (2) in the Teflon-coated spinneret, the Teflon tube's edge is slightly higher than the two metal capillaries, thereby enabling the two fluids to converge first and then flow to the electrical field to ensure a matching step of drawing under the electrical field and thus integrate the Janus nanofibers, as shown in the diagram of Figure 8e and the digital picture in Figure 8f; and (3) in electrospinning, the reasonable design of the spinneret not only provides a template for creating nanostructures but should also be safe and convenient for usage. Aside from these aspects, the spinneret should obey the fundamental rules for guiding and organizing the working fluids into the electrical fields in a proper manner.

From Safety Education to Innovations for Collecting Nanofibers

All the above-mentioned examples pertain to innovations for the novel spinneret, which initially stemmed from safety education on avoiding electrical shocks. Furthermore, from safety education on the operation, students can effectively overcome their psychological fear of high voltage, thereby realizing innovations for collecting electrospun nanofibers. Two examples are presented in Figure 9.

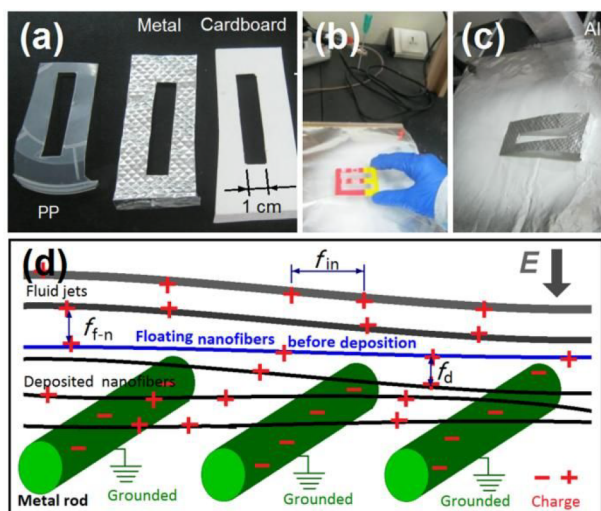


Figure 9. From safety operation education to innovations for collecting nanofibers: (a) three grooves made of polypropylene, aluminum foil, and cardboard; (b,c) cardboard groove and aluminum foil grooves are utilized to collect nanofibers for sampling TEM; (d) a diagram showing that parallel metal rods can be utilized for depositing nanofibers.¹⁷

The sampling for TEM analyses is not an easy task for the students because the thickness of the sample for observation must be controlled to be very thin, often less than 80 μm . Some students are afraid of the electrospinning system and always stand far away from the apparatus while wearing two layers of gloves. However, the real danger is the high electric

current. Although the applied voltage is 10 kV, often the electric current is in the range of nanoamperes. Thus, we do not need to be afraid of the equipment if appropriate protective measures are in place.

Based on safety education, the students conceived two innovations related to the collection of nanofibers. First, they can stretch their hand under the electric field. As shown in Figure 9a, they prepared three grooves with different textures: polypropylene, aluminum foil, and cardboard. They utilized these grooves to collect nanofibers (Figure 9b,c shows the aluminum foil and cardboard grooves, respectively). The copper grid with a carbon membrane can be passed through the grooves to prepare nanofiber samples for TEM observation. Having overcome their fear because of safety education, the students can apply this innovation to the collector. Figure 9d shows a diagram of parallel metal rods that can be utilized to collect aligned nanofibers.¹⁷ During the electrical drawing processes, the nanofluids are elongated by the repelling of born charges in the fluids (f_{in}) in the electrical field (E). Meanwhile, the solvents are removed to solidify the fluids into nanofibers. Repelling forces exist between the fluid and solidified nanofibers (f_{f-n}) and between the deposited nanofibers and the floating ones (f_d). When a floating nanofiber is deposited on the collector, it should be subjected to a series of forces from its surroundings. These forces include the repulsion forces from the already deposited nanofibers (f_d), the repulsion forces from the fluid jets that are still in the bending and whipping coils (f_{f-n}), and the forces from the electrical field (i.e., the repulsion forces from the positive electrode and the attractive forces from the negative electrode). Under the same strength of the electrostatic field, the higher the density of charges of the fluid jets and deposited nanofibers, the larger the forces of f_d and f_{f-n} , and the better uniaxial effect with which the floating nanofibers align. Under the same applied voltage, the polymer-coated spinneret can prevent electrical energy loss to the environment effectively to forward a higher density of charges to the fluid jets as compared to the metal spinneret, thereby achieving a large repelling force and a better alignment effect. Safety education and the innovations for collecting nanofibers can deepen students' cognition of the nanofabrication mechanism of electrospinning and other EHDA methods, such as electro-spraying and 3D printing. The scale of chemical events is small, and we need to contemplate the science using our imagination.⁴⁸ Here, the safety issues and the related theories can help to cultivate the students' imagination. By the way, the electrospinning and electrospun nanofibers are abundant in providing useful teaching materials for engineering education and science education.^{49,50}

After several years' laboratory safety and practice education, the student learning effects are significantly improved. These improvements are reflected at least in the following aspects: (1) the number of students in the course was increasing from 17 for 2020, to 80 for 2021, and to 89 for 2022; (2) the average scores of the students increased year after year from 85.12, to 89.33, and to 90.85; (3) more and more students exploited electrospinning to begin their innovation training programs; (4) the number of students that selected electrospinning as a topic to conduct their graduation projects was always increasing; (5) the scientific publications (supervised by Deng-Guang Yu) from both postgraduate and undergraduate students increased from 15, to 20, to 26 for 2020, 2021, and 2022, respectively. Some junior or senior undergraduate

students published their maiden works in academic journals,^{51–53} achieved some innovation competition awards, and are keen on a hand-held electrospinning apparatus.³⁵ These outcomes are impossible without the laboratory safety education for the students.

CONCLUSIONS AND OUTLOOKS

In this article, electrospinning, which is a popular process for creating polymeric nanofibers, is explored as an example to encourage students to conduct technique innovations on the working processes from a special standpoint, i.e., safety education on laboratory experiments. Although electrospinning is a popular technique for creating nanomaterials, it has a series of safety issues during its application. In addition to the common safety issues, such as chemical and physical safety risks, special safety issues are related closely to its implementations. Safety education on how to avoid electric shocks on the spinneret has encouraged students to realize a series of innovations on the spinnerets and the related electrospinning processes. These innovations include modifications to the single-fluid blending monoaxial, two-fluid coaxial, trifluid triaxial, and side-by-side processes. Safety education also helps students overcome their fear of high voltage while performing experiments on electrospinning and applying innovations to the collection of nanofibers at the collectors. Thus, effective safety education not only ensures safe, energy-saving, and convenient implementations of different electrospinning processes but also helps deepen and broaden students' professional knowledge on electrospinning and stimulates their enjoyment of technological innovation.

Whether during graduate education, undergraduate education, or specialized and vocational education, students should be only imparted with not only new knowledge but also the "what, why, and how," with the "how" being the most important, because society needs skilled workers and professional technicians with technical know-how in professional fields. Safety education can be combined with theoretical and practical "know-how" education to nurture the students, although becoming professionals and technicians is never easy. For all new professional techniques, safety education is a special aspect that should be fully explored to promote the technological innovations of students in HEIs and in turn improve their capability to integrate theories with practices for more innovations.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available at <https://pubs.acs.org/doi/10.1021/acs.jchemed.2c00568>.

Instructor Notes - Detailed information about (1) the trend of electrospinning and its related scientific fields; (2) experiments about how to set up a homemade electrospinning system; and (3) experiments about how to implement a coaxial electrospinning process are designed as a document for instructors ([PDF](#), [DOCX](#))

Post survey ([PDF](#), [DOCX](#))

AUTHOR INFORMATION

Corresponding Authors

Deng-Guang Yu – School of Materials and Chemistry, University of Shanghai for Science and Technology, Shanghai

200093, China; orcid.org/0000-0001-7825-4498;

Email: ydg017@usst.edu.cn

Tao Zhou – School of Materials and Chemistry, University of Shanghai for Science and Technology, Shanghai 200093, China; Email: zhoutao310@163.com

Authors

Qiang Li – School of Materials and Chemistry, University of Shanghai for Science and Technology, Shanghai 200093, China

Wenliang Song – School of Materials and Chemistry, University of Shanghai for Science and Technology, Shanghai 200093, China; orcid.org/0000-0002-8005-5460

Lin Xu – School of Materials and Chemistry, University of Shanghai for Science and Technology, Shanghai 200093, China

Kaihui Zhang – School of Materials and Chemistry, University of Shanghai for Science and Technology, Shanghai 200093, China

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.jchemed.2c00568>

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This study is financially supported by the Shanghai Natural Science Foundation (No.20ZR1439000), Graduate course Ideological and political research project in USST & DSDG education project.

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